

Computer Networks

Lecture 3: OSI Model

Content



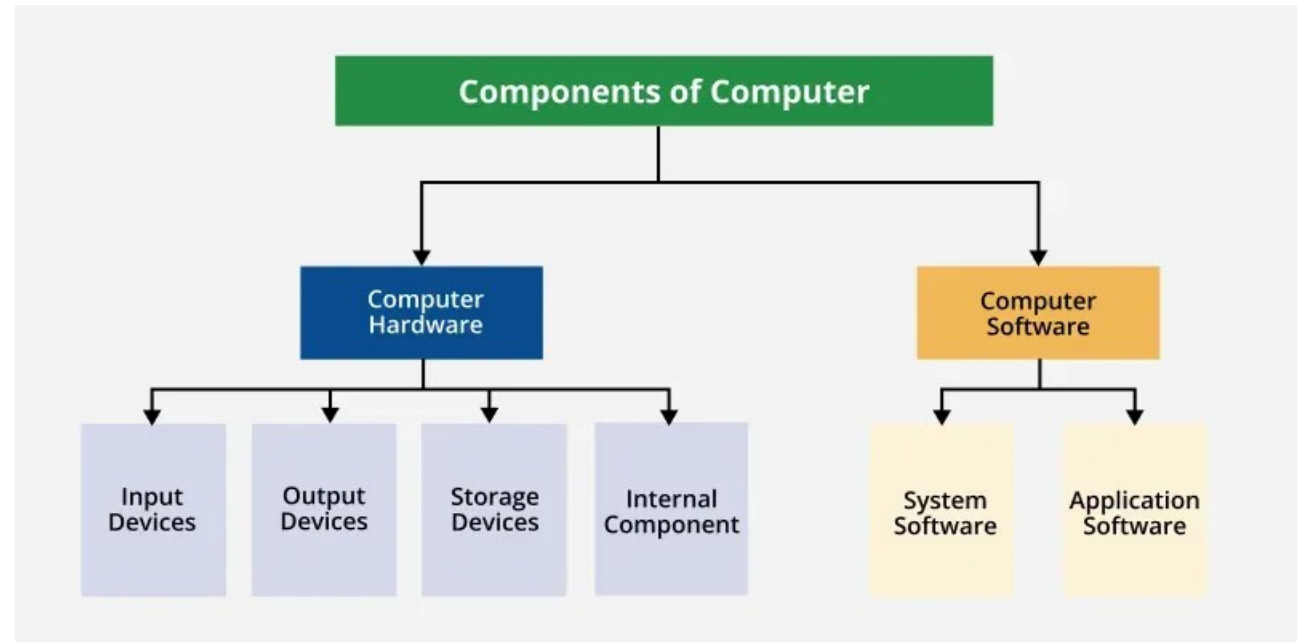
- **Network Delays**
- **OSI Model**

Network Hardware and Software

A network is a combination of hardware and software that sends data from one location to another.

Hardware consists of physical equipment that carries signals across the network.

Software consists of instruction sets that enable the services expected from a network.

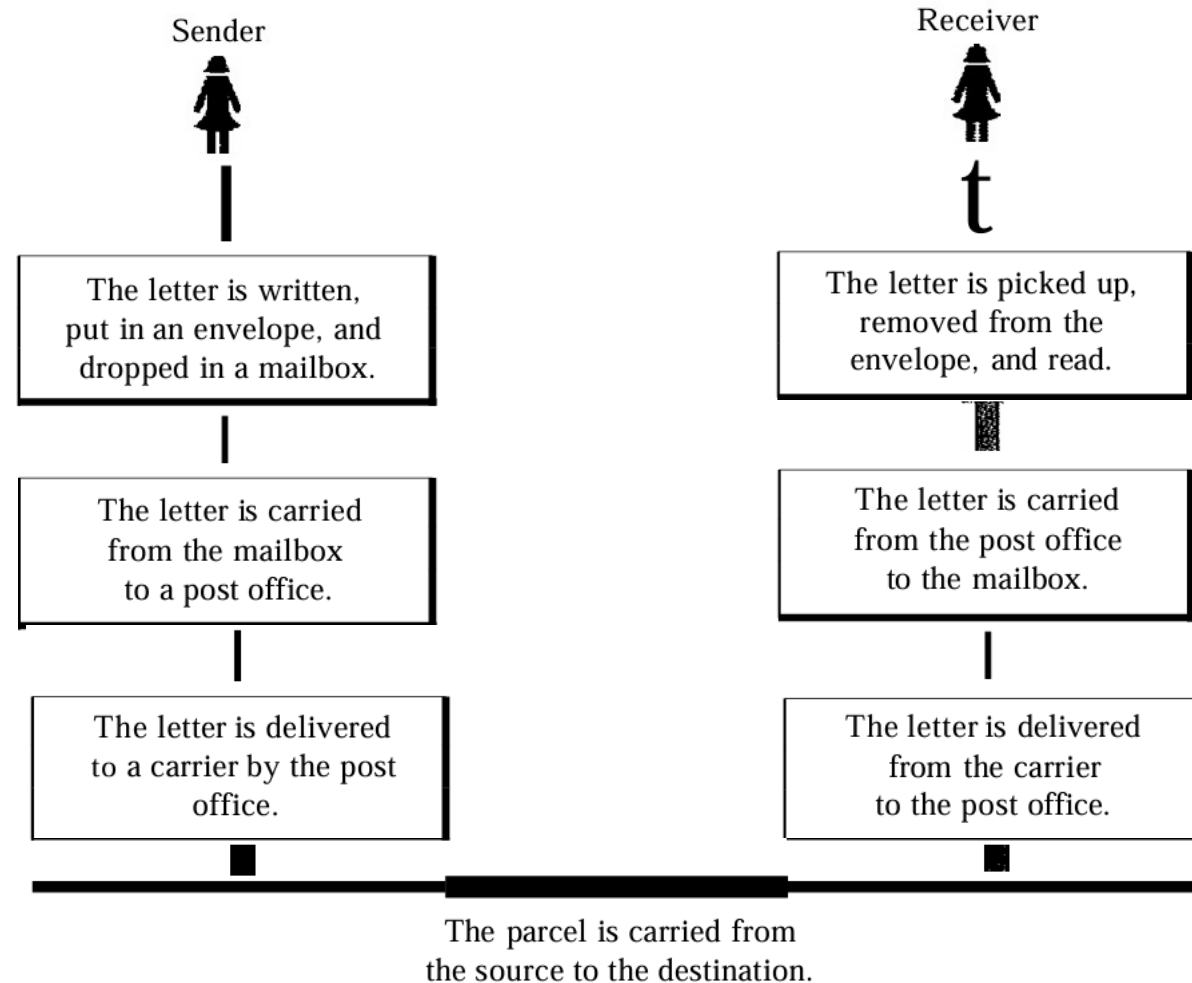


Using only hardware makes communication tasks complex and inefficient.

Layered Communication

Software simplifies networking by dividing communication into manageable tasks.

Example: The process of sending a letter



Internet

Internet is worldwide network of computer networks.

How does it work?

- ✓ Most computers are not connected directly to the internet.
- ✓ They are connected to smaller networks.
- ✓ Which are connected through gateways to the internet backbone.

Network Model

Definition

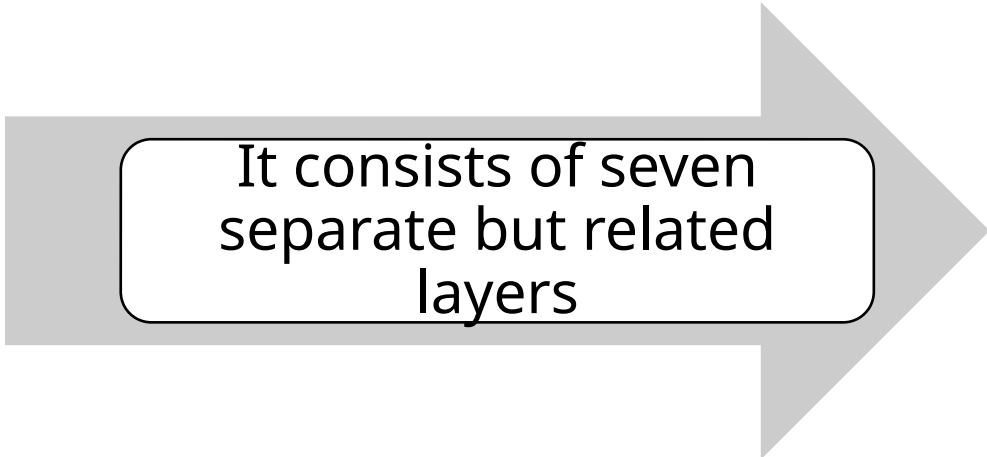
A method of describing and analyzing data communication networks by breaking the entire set of communication process into a number of layers.

International standard organization (ISO) established a committee in 1977 to develop an architecture for systems communication.

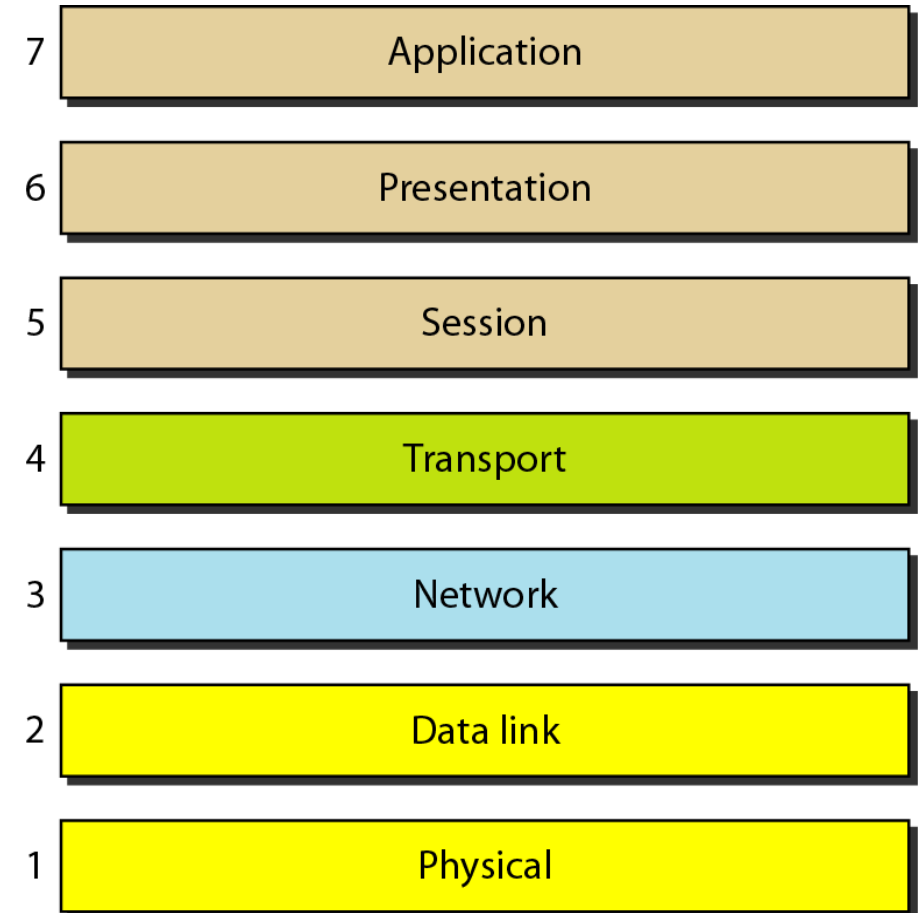
- Open System Interconnection (OSI) reference model is the result of this effort.
- This model allows any two different systems to communicate regardless of their underlying architecture

OSI Model

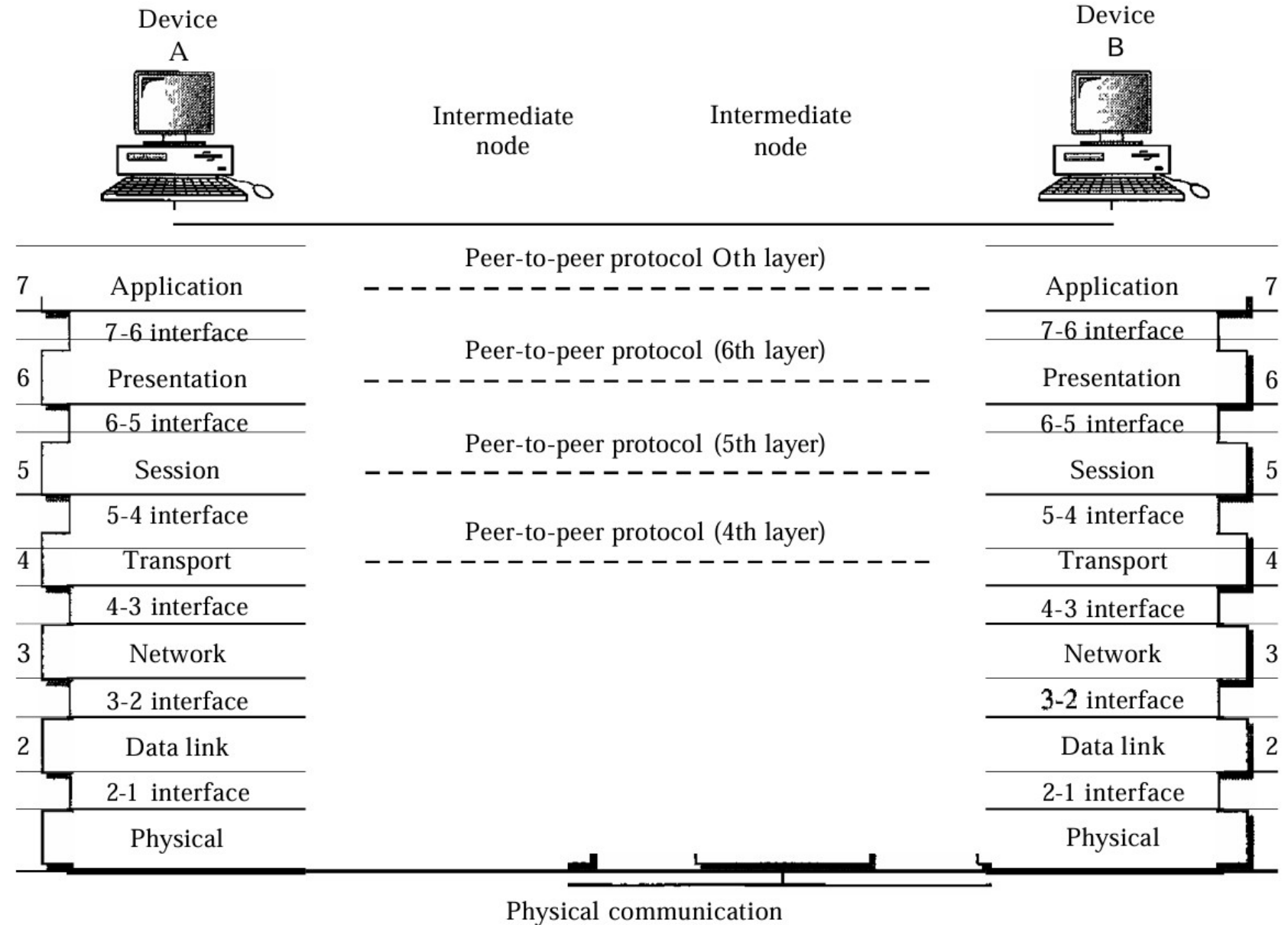
The OSI model is a layered framework for the design of network systems that allows communication between all types of computer systems.



It consists of seven
separate but related
layers



Layer interactions in OSI model

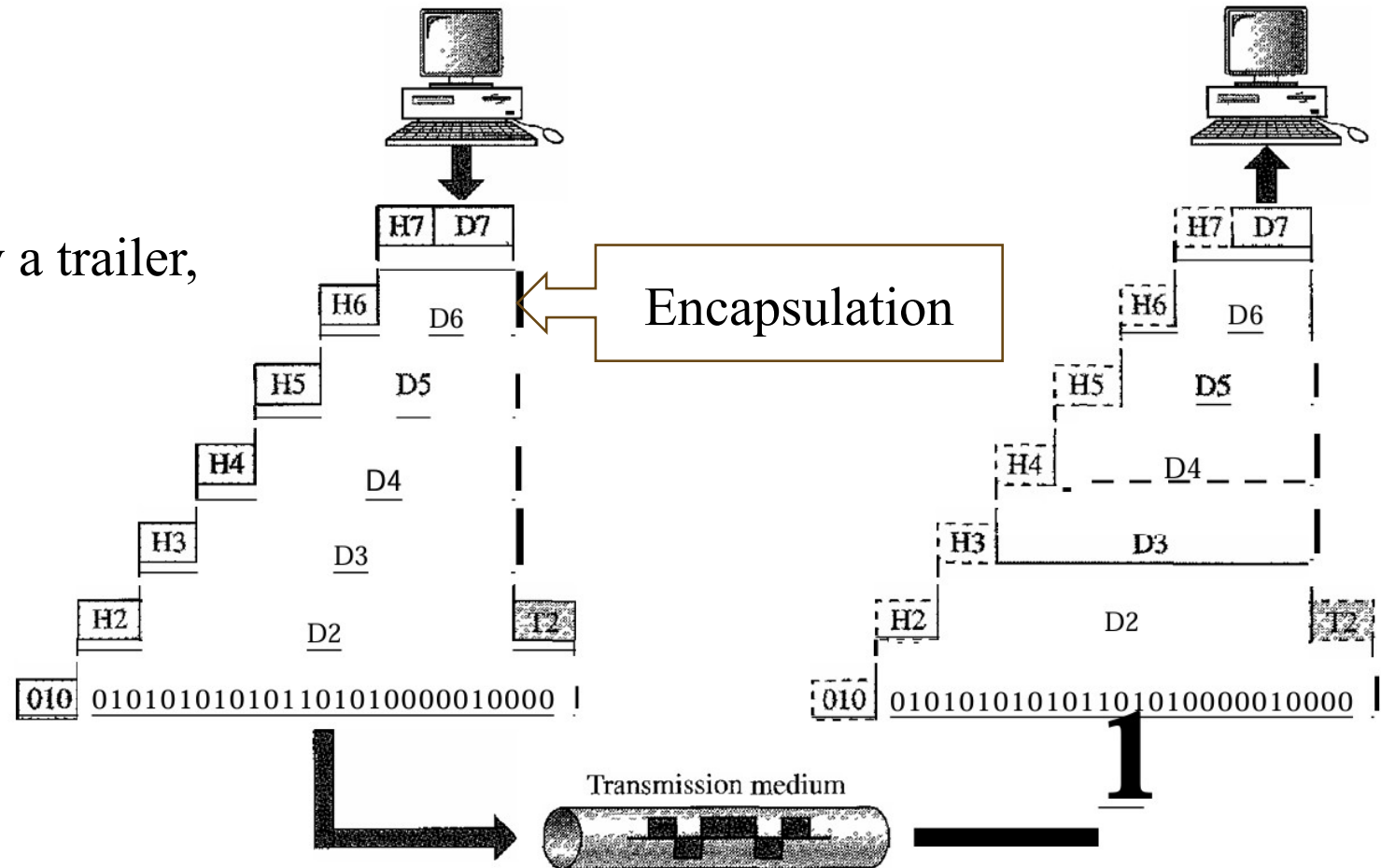


Layer interactions in OSI model

- Layers 1, 2, 3- physical, data link and network are network support layers.
- Layer 4, the transport layer, links the two subgroups.
- Layers 5, 6, 7- session, presentation, and application are user support layers.

Using OSI Model

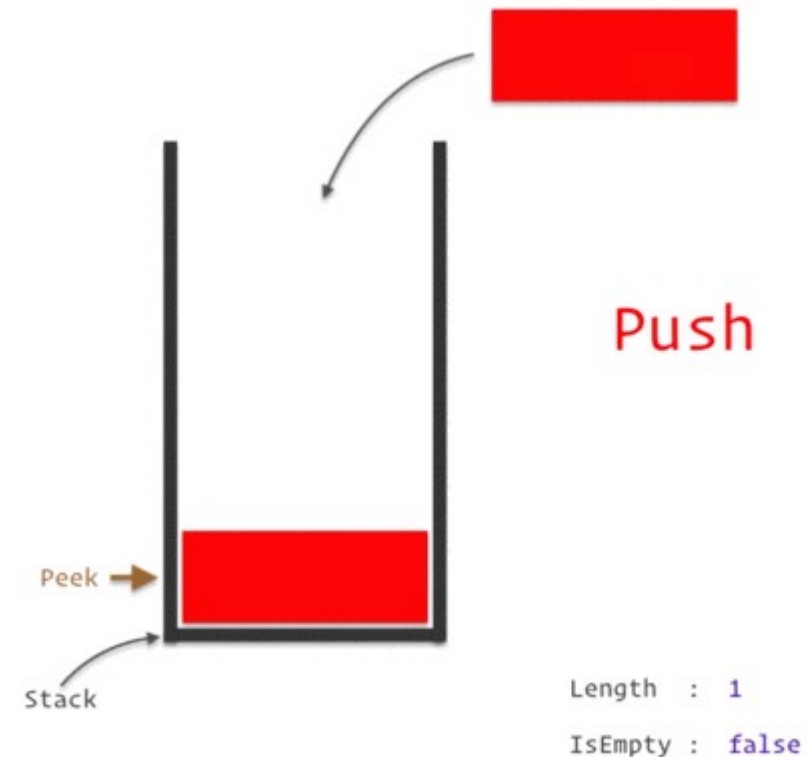
At each layer, a header, or possibly a trailer, can be added to the data unit.



Protocol Stack

At sender we add header but at receiver we remove header

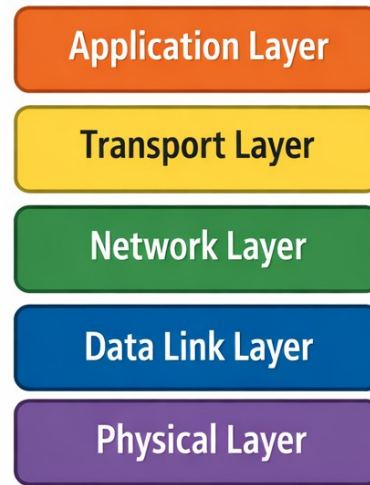
This architecture is called protocol stack architecture because like stack the 1st header added is the one which is removed at the last or the last added header is the one which is removed first



Delays in computer network

Time taken to place the data on channel is transmission time/transmission delay.

The data has to be propagated to reach the destination this time is called as propagation time/propagation delay.



Transmission Time

Time to place all bits onto the wire

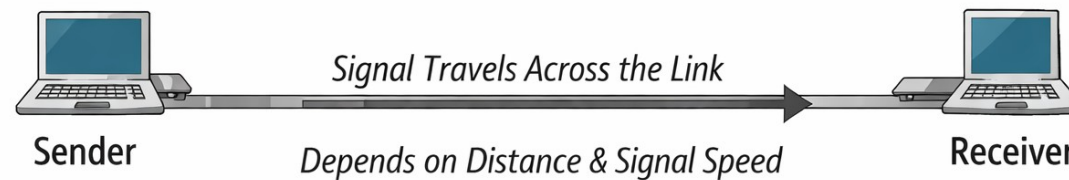


Depends on Link Bandwidth

$$\frac{\text{Message Size}}{\text{Bandwidth}}$$

Propagation Time

Time for signal to travel from sender to receiver



$$\frac{\text{Distance}}{\text{Velocity}}$$

Delays in computer network

Processing Delay/ Processing Time

Processing delay is the time taken by a network device (such as a router or switch) to process a received packet.

Queuing Delay/ Queuing Time

Queueing delay is the time a data(packet) spends waiting in a queue (buffer) before it can be processed or transmitted. It varies depending on network traffic conditions.

$$\text{Total Delay} := T_t + T_p + T_q + T_{pro}$$

Where:

- T_t = Transmission Delay
- T_p = Propagation Delay
- T_q = Queueing Delay
- T_{pro} = Processing Delay

Generally processing time and queuing time are ignored as they are very small and the effective delay or time is considered as

$$\text{Total Delay} := T_t + T_p$$

Question

1. A message of size 1 KB is transmitted over a communication link with a bandwidth of 1 Mbps. Calculate the transmission time required to send the message.
2. If the distance between sender and receiver is 3KM and velocity is 2×10^8 . Calculate the propagation time

Delays in computer network

When the data is transmitted, acknowledgement is also expected from the receiver end so that we can ensure that transmission is completed.

Thus the total time taken to complete the communication (obviously for one packet) is

$$\text{Total Time: } T_{t.data} + T_{p.data} + T_{t.ack} + T_{p.ack}$$

As size of acknowledgement is very small thus the transmission time of acknowledgement is ignored and the propagation time for both data and acknowledgement is assumed to be same

$$\text{Total Time: } T_t + 2 \times T_p \Rightarrow T_t + R.T.T.$$

Link Utilization

As sender has only utilized the transmission time and after that sender is only waiting for acknowledgement thus the link utilization for the sender is given by

$$\text{Link Utilization for sender} = \frac{T_t}{T_t + 2 \times T_p} \Rightarrow \frac{T_t}{T_t + R.T.T}$$

Question

1. If the link utilization is to be 50%, and given that the propagation time is 10 milisec. Calculate the Transmission time.
2. If the link utilization is to be 75%, and given that the R.T.T is 10 milisec. Calculate the Transmission time.
3. If the message size is L and bandwidth is B . Also given that the R.T.T. is R ,
 - a. Calculate the link utilization for the sender.
 - b. What should be the message size for 50% link utilization.